

Deep Learning - Assignment 1

Top 10 Real World Applications of Deep Learning

Name: Srinidhi R

USN: 4GW23CI053

Subject: Deep Learning

Assignment: 1

No	Application
1	Medical Diagnosis & Healthcare
2	Self-Driving Cars
3	Image Recognition & Computer Vision
4	Natural Language Processing (NLP)
5	Speech Recognition
6	Fraud Detection in Banking
7	Recommendation Systems
8	Drug Discovery
9	Game Playing AI (AlphaGo)
10	Deepfake & Image Generation

Note: I have taken Fraud Detection in Banking (Sl. No. 6) for the detailed study below.

Elaborate Study: Fraud Detection in Banking Using Deep Learning

Introduction

Banks lose a huge amount of money every year because of fraud like stolen cards, fake identities and account hacking. Earlier, banks used simple rule based systems, for example "block the transaction if amount is more than Rs.50,000". But fraudsters keep changing their methods so these fixed rules fail after some time. Deep Learning solves this problem because it can learn the hidden patterns of genuine and fraud transactions directly from large amount of data, and it keeps adapting itself as new types of fraud appear.

How It Works

The deep learning model is trained using past transaction data which is already labelled as "genuine" or "fraud". The model learns to find the combination of features which usually means a transaction is suspicious. Some of the features used are:

- Transaction amount and merchant type
- Location of transaction and device/IP used
- Time of transaction and how fast transactions are happening one after another
- Customer's normal spending pattern
- Sequence of previous transactions

The models mostly used are Artificial Neural Networks (ANN) for transaction data, LSTM (a type of RNN) for looking at spending pattern over time, Autoencoders which can detect fraud even without labelled data (unsupervised), and Graph Neural Networks (GNN) which are used to catch fraud rings by studying the connections between accounts and devices.

Working Process

1. Transaction data is taken as input (amount, location, device, time, merchant)
2. Data is cleaned and normalized (preprocessing)
3. Data is passed to the Deep Learning model (ANN / LSTM / Autoencoder / GNN)
4. Model extracts patterns and finds anomalies
5. Model gives a fraud probability score
6. Based on score, transaction is approved, flagged or blocked in real time

Real World Examples

Application	Technology Used
Credit Card Fraud Detection	ANN with Gradient Boosting
Real-Time Payment Screening	LSTM on transaction sequence
Anti-Money Laundering (AML)	Graph Neural Networks
Account Takeover Detection	Autoencoders (unsupervised)
Insurance Claim Fraud	Deep Neural Networks

Famous Systems Using DL for Fraud Detection

- PayPal - uses deep learning on billions of transactions to reduce fraud losses
- Mastercard Decision Intelligence - scores every transaction in real time using AI
- Visa Advanced Authorization - neural network scoring in milliseconds

- HSBC - uses AI for anti-money laundering and to reduce false alarms
- Feedzai - deep learning platform used by several banks for fraud and risk scoring

Advantages

- Can detect fraud in real time, within milliseconds
- Learns new fraud patterns automatically without changing rules manually
- Gives fewer false alarms compared to old rule based systems
- Can handle huge number of transactions happening across the world
- Can find hidden fraud rings using graph based models

Challenges

- Fraud cases are very less compared to genuine ones, so data is imbalanced
- Fraud patterns keep changing, so model needs to be retrained often
- Deep learning models are like a "black box", hard to explain why a transaction was flagged
- Prediction has to happen very fast (real time), which is technically difficult
- Data privacy rules like RBI guidelines and GDPR need to be followed

Simple Code Example

```
import tensorflow as tf
from tensorflow.keras import layers, models

model = models.Sequential([
    layers.Dense(64, activation='relu', input_shape=(30,)),
    layers.Dropout(0.3),
    layers.Dense(32, activation='relu'),
    layers.Dense(1, activation='sigmoid')
])

model.compile(optimizer='adam',
              loss='binary_crossentropy',
              metrics=['accuracy'])

model.fit(X_train, y_train, epochs=20, batch_size=256)
```

Conclusion

Deep Learning has made fraud detection in banks much faster and smarter than before. Instead of depending on fixed rules, banks can now use models which keep learning from new data and adapt to new fraud tricks. This helps in protecting customers' money and also reduces the loss for banks. Still there are challenges like explainability and data imbalance which

researchers are working on to make these systems even better.

References

- Mastercard Decision Intelligence - mastercard.com
- PayPal Engineering Blog - paypal.com
- Feedzai - feedzai.com
- Deep Learning by Ian Goodfellow, Yoshua Bengio, Aaron Courville
- IEEE Transactions on Neural Networks - Fraud Detection Survey